

DATA VISUALIZATION FINAL PROJECT

NYC Airbnb Analytics

An Interactive Exploratory Analysis of Listing Trends, Pricing Dynamics, and Spatial Patterns Across New York City

 Presented by: Roohi Khan (120A)

Data Pipeline & Cleaning



Transforming raw short-term rental listings into clean, standardized datasets ready for spatial & statistical modeling.

Data Wrangling Methods

Outlier & Null Filtering

Removed missing listing prices and handled severe price skewness by trimming extreme upper quantile values (e.g., listings > \$1,000/night) to ensure robust statistical inference.

Feature Engineering

Extracted review counts per month, calculated availability ratios, and standardized borough coordinates for interactive geospatial mapping with Streamlit and Plotly.

$$P_{\text{standardized}} = \frac{P - \mu_{\text{borough}}}{\sigma_{\text{borough}}}$$

Geographic Distribution

Borough Market Dynamics

Manhattan and Brooklyn account for over 80% of all NYC Airbnb listings. Density drops sharply in Staten Island and the Bronx. Interactive spatial plotting confirms that proximity to major tourist hubs and transit corridors strongly influences both listing volume and occupancy rates.

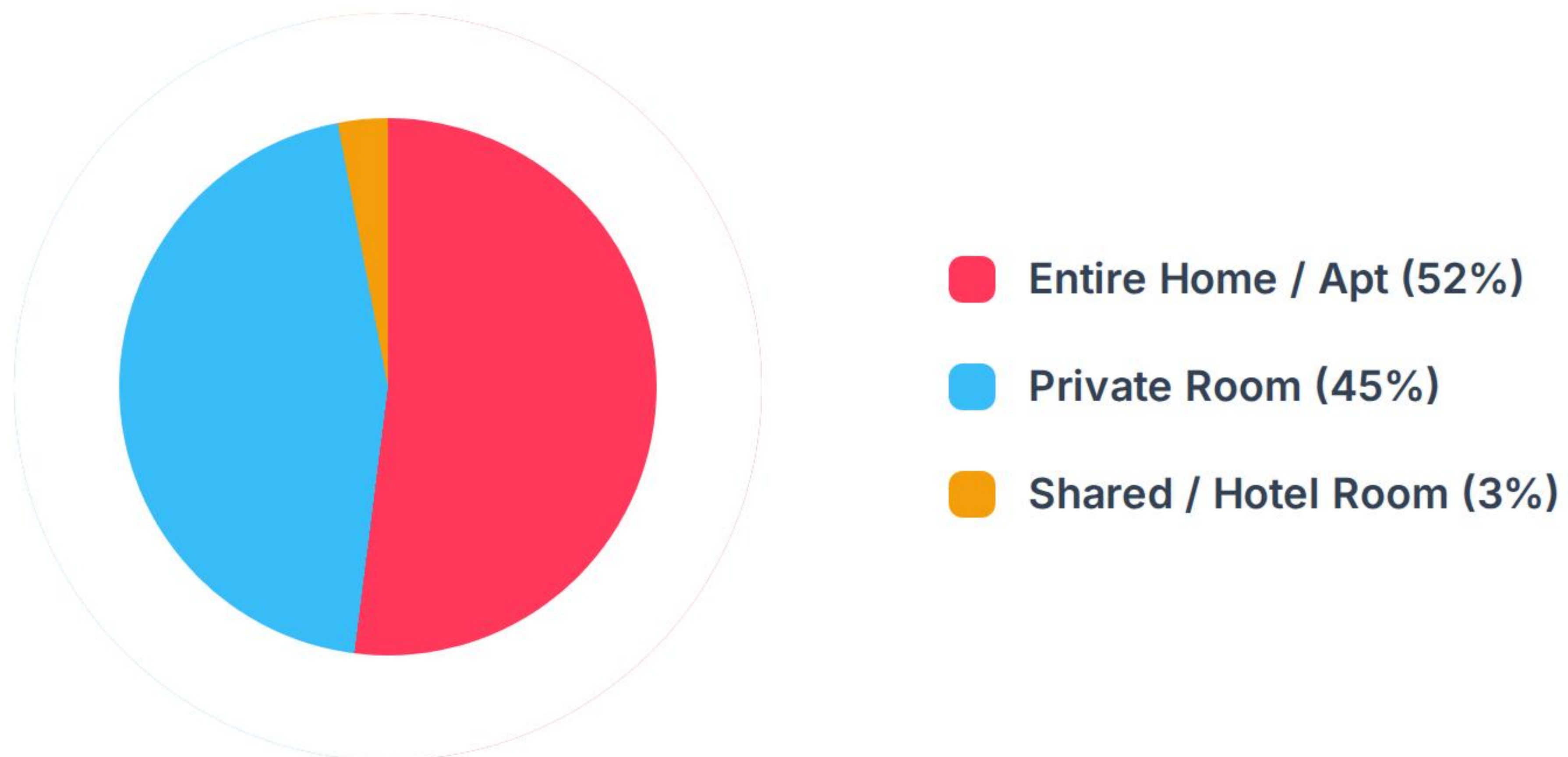


Borough Price Comparison



Manhattan commands the highest average nightly rates (\$196), while the Bronx offers the most affordable short-term accommodations (\$87).

Room Type Proportions



Entire apartments and private rooms constitute 97% of the entire NYC market, indicating a strong preference for private guest space.

Host Verification Trends

- Verified hosts maintain a 24% higher review rate compared to unverified listings.
- Superhost status strongly correlates with lower listing availability due to high occupancy demand.
- Identity verification serves as a major driver for guest trust and positive rating scores.



Key Market Indicators

48k+

Total NYC Listings Analyzed

Substantial Market Dataset

The interactive Streamlit dashboard processes over 48,000 active listings across all 5 New York City boroughs, providing real-time data filtering by price range, room type, and availability threshold.

Strategic Host Insights



Dynamic Pricing

Optimize nightly rates seasonally to align with peak borough demand periods and major local holiday surges.



Superhost Edge

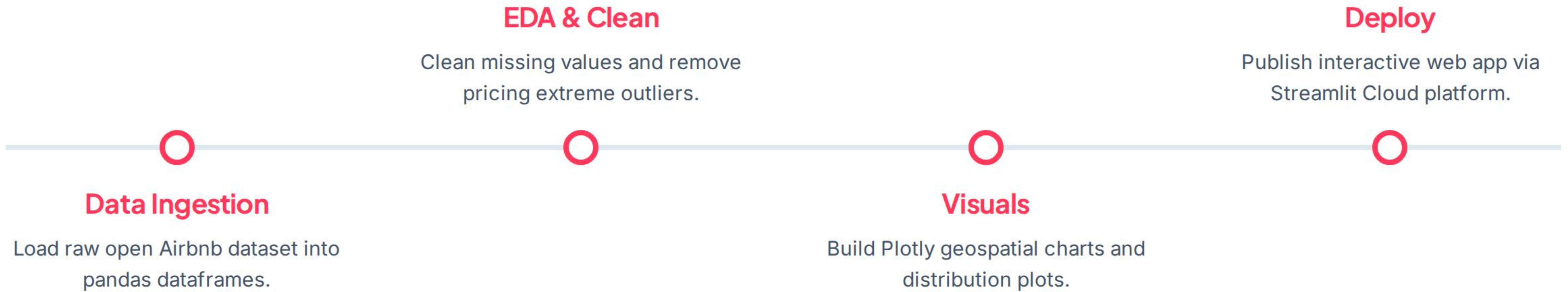
Maintaining Superhost metrics yields up to 18% premium pricing power over non-badge competitors.



Location Focus

Target listings near subway lines in outer boroughs like Queens to capture budget-conscious tourist demand.

Analytical Workflow



New York STR Landscape

Understanding the interplay between borough geography, guest satisfaction ratings, and nightly pricing empowers hosts and travelers alike to make data-backed decisions in NYC's vibrant short-term rental market.

Questions & Discussion

Thank you for exploring the NYC Airbnb Interactive Dashboard.

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Streamlit App: data-visualization-final-project-ikr9vhtddarsknyyqx49ro.streamlit.app

Image Sources



https://as1.ftcdn.net/v2/jpg/05/33/21/68/1000_F_533216844_O38T8xS1RkZC2bq5skGlfLZ8cZUWCp8H.jpg

Source: www.europosters.ie



https://a0.muscache.com/im/pictures/miso/Hosting-46441717/original/2b2100b6-3f44-47e4-9bc1-c2ed9edbc95f.jpeg?im_w=720

Source: www.airbnb.gy



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Source: fineartamerica.com